Kubernetes-Native AI Repair Loop

Automated Failure Detection and Self-Healing
for Confidential GitOps Pipelines Using SCONE

Standard Kubernetes Failures

- `CrashLoopBackOff` / `OOMKilled`
- `ImagePullBackOff`
- Pending on SGX-capable nodes
- Argo Workflow stage failures

Visible via Kubernetes API – but need AI triage

Confidential Computing Failures

SGX rejections · MRENCLAVE mismatches · CAS session errors · Enclave startup faults · Missing env-var injections

INVISIBLE to standard Kubernetes tooling

Confidential GitOps pipeline
SCONE · SGX · Argo · Kubernetes

Visible failures	Hidden failures
`CrashLoop`	SGX rejection
`OOMKilled`	MRENCLAVE err
Node Pending	CAS error
Stage failure	Enclave fault
↓ kubectl sees	↓ kubectl misses

THE GAP
No tool addresses both failure classes simultaneously

Confidential pipeline failures are diagnosed **manually** — slow, expert-dependent, and unsustainable at scale.

↑ Scale

- Each image adds CAS session, MRENCLAVE, attestation flow
- Every deploy = new failure points

○ Cost

- SGX, CAS, enclave errors invisible to `kubectl`
- Hours spent per failure

⋆ Opportunity

- AI repair works for standard pipelines – extend to SGX
- SCONE API + SGX cluster already available

3. Research Gap & Existing Work

Validated against published literature

Paper	Contribution	Gap	Our use
LogSage ASE 2025 [1]	LLM CI/CD repair. 98% precision; 1M+ runs	Standard only; no SGX/CAS; no TEE	Primary baseline
Self-healing K8s 2025–26 [2]	LLM+RL autonomous K8s recovery	Standard K8s only; blind to confidential class	Architecture reference
SAP HANA LLM 2025 [3]	Domain prompts: 84% → 92% accuracy	No K8s; no attestation diagnostics	Informs RQ2 prompt design
SCONE Conf. AI Jan 2026 [4]	Conf. AI on K8s; <5% overhead	Infrastructure only; no repair loop	We build the loop on top

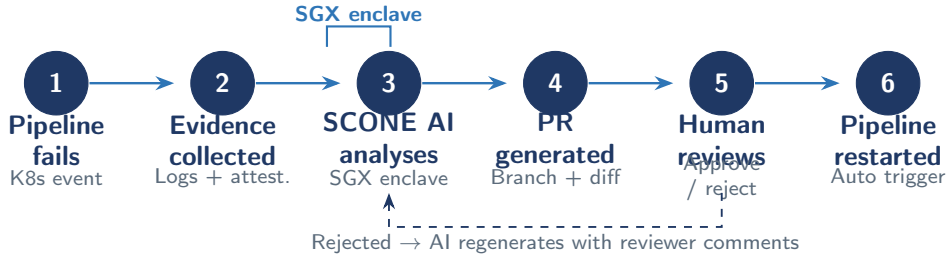
Confirmed gap: No system combines (1) K8s failure detection, (2) SGX/CAS diagnostics, (3) AI in a TEE, (4) automated PR repair

SCONE Confidential AI API · SGX Kubernetes cluster · SCONE CAS access

#	Area	What we build
01	K8s evidence collection	Pod logs, events, `CrashLoopBackOff`, `OOMKilled`, SGX node status via Kubernetes API
02	Attestation diagnostics	SGX rejections, `MRENCLAVE` mismatches, CAS session errors – invisible to standard tooling
03	Prompt engineering	Structure evidence into prompts for SCONE AI accuracy [3]
04	Classification & patching	Map AI output to 1–2 line patch: `scone.yaml`, `transform.sh`, `assumptions.md`
05	Automated PR generation	Commit fix, open PR, pre-commit hooks, auto re-trigger on merge
06	Recurrence tracking	Track patterns across namespaces; promote to shared templates when threshold ($\geq 3\times$) met

5. Repair Loop Architecture

End-to-end flow – failure to self-healing



Key differentiator: Step 3 runs inside a SCONE-protected SGX enclave. Confidential diagnostics never leave hardware-encrypted memory. The host OS and Kubernetes control plane cannot read them.

6. Research Questions

Primary RQ: How can a Kubernetes-native AI repair loop using SCONE confidential AI automatically detect, classify, and fix confidential GitOps pipeline failures?

RQ	Area	Question
RQ1	Evidence collection	What K8s events + SCONE diagnostics maximise failure evidence completeness?
RQ2	Prompt engineering	What format produces highest SCONE AI API classification accuracy?
RQ3	Accuracy	How accurately does the SCONE AI classify failures across both classes?
RQ4	Latency	What is the end-to-end latency from K8s event to open PR?
RQ5	Effectiveness	What proportion of AI-generated PRs fix the failure on re-execution?

References I

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